Your grade: 80%

Your latest: **80%** • Your highest: **80%** • To pass you need at least 80%. We keep your highest score.

Instructions

This quiz will use all of the Coursera Lab Jupyter notebooks from this week. I recommend that you have another browser tab or window available and that you open the labs there. You can run the labs in parallel with the quiz, and enter quiz answers from the labs directly.

1. To compute $\hat{x}_k^-$, $\Sigma_{\tilde{x},k}^-$, and $\hat{z}_k$, what information do we require? Select all that apply. — 1 / 1 point

- [x] We need to know model matrices A_{k-1}, B_{k-1}, C_k, and D_k.

  Correct
  Yes. These matrices are needed in the prediction steps.

- [x] We need to know the process-noise covariance matrix $\Sigma_{\tilde{w}}$.

  Correct
  Yes. We need to know this information for step 1b.

- [ ] We need to know the sensor-noise covariance matrix $\Sigma_{\tilde{v}}$.
- [x] We need to have an estimate of the prior values of the state vector, $\hat{x}_{k-1}^+$.

  Correct
  Yes. This is needed in step 1a.

- [x] We need to know the error covariance matrix of the prior state estimate, $\Sigma_{\tilde{x},k-1}^+$.

  Correct
  Yes. This is needed in step 1b.

2. What is the significance of the state estimation error covariance matrix $\Sigma_{\tilde{x},k}^+$? — 1 / 1 point

- ( ) It indicates the uncertainty of the predicted state.
- ( ) It is needed to compute the output prediction error $\tilde{z}_k = z_k - \hat{z}_k$.
- ( ) It is needed to be able to compute the Kalman gain matrix L_k.
- (•) It allows computing error bounds on the state estimate produced by the Kalman filter.

  Correct
  Yes. This is the primary significance and use of this matrix.

Instructions for the next questions.

3. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.2] — 1 / 1 point

What is the true velocity of the mass at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

Hint: Time 25 seconds corresponds to `k = 251` in the Octave code.

0.055395

Correct
Yes. That is correct.

4. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.2] — 1 / 1 point

What is the measured position $z(t)$ of the mass at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

0.073309

Correct
Yes. That is correct.

Instructions for the next questions

5. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.3] — 0 / 1 point

What is the position-estimation error at time $t = 25$ seconds? Recall that error is a signed value and is calculated as "true position minus estimated position". Enter your answer with four digits to the right of the decimal point.

0

Incorrect
No. Make sure that you are computing error correctly and that you are reporting the position error and not the velocity error.

6. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.3] — 0 / 1 point

What is the 3σ bound on the position error at time $t = 25$ seconds? Enter your answer with four digits to the right of the decimal point.

0.0074352

Incorrect
No. Make sure that you are reporting position bounds and not velocity bounds.

Instructions for the next questions

7. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.4] — 1 / 1 point

Run the open-loop state-estimator code from the Coursera Lab. What is the 3σ error bound on the position state at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

0.058301

Correct
Yes. That is correct.

8. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.4] — 1 / 1 point

Run the Kalman-filter state-estimator code from the Coursera Lab (for the case where the initial state is poorly chosen). During the simulation, how many times did the magnitude of the position error exceed the 3σ bounds on position error?

0

Correct
Yes. That is correct.

Instructions for the next questions

9. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.5] — 1 / 1 point

Run the code cells in the notebook from the section titled: "Implementing a Kalman filter using incorrect noise variances". What is the fraction of time that the velocity-estimate error is outside its 3σ error bounds? Enter your answer with two digits to the right of the decimal point.

Hint: You can compute this answer using the following line of code:

```
length(find(abs(xerr(2,:))>boundBVstore(2,:)))/nt
```

0.31269

Correct
Yes. That is correct.

10. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.5] — 1 / 1 point

Run the code cells in the notebook from the section titled: "Implementing a Kalman filter with correlated noises". What is the fraction of time that the velocity-estimate error is outside its 3σ error bounds? Enter your answer with three digits to the right of the decimal point.

0.24975

Correct
Yes. That is correct.